

EXPERIMENT 7

Student Name: Jay Varshney

UID: 24BDA70206

Branch: AIT-CSE

Section/Group: 24AIT_KRG-G2

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Experiment 7 – Implement Router as DHCP server that can serve multiple VLAN's.

Aim: To implement Virtual Local Area Networks (VLANs) and VLAN trunking on network switches, and to configure communication between different VLANs using appropriate trunking protocols.

Objective

- To understand the concept and purpose of VLANs in a computer network.
- To create and configure multiple VLANs on a network switch.
- To assign switch ports to specific VLANs.
- To configure trunk links for carrying traffic from multiple VLANs.
- To understand the operation of VLAN trunking protocols such as IEEE 802.1Q.
- To verify connectivity and VLAN communication using suitable network commands.

Theory:

VLAN (Virtual Local Area Network)

A VLAN is a logical network created within a physical switch. Instead of putting every device connected to a switch into the same logical network, devices can be divided into different VLANs. For example, VLAN 10 can represent Students, while VLAN 20 can represent Faculty. Although the devices may be connected to the same physical switch, VLANs logically separate them.

Need for VLANs

VLANs are used for network segmentation, security, broadcast domain reduction, better network management, and logical grouping of users.

Inter-VLAN Routing

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Different VLANs are separate Layer 2 networks, and a Layer 2 switch cannot normally route between them. Communication between different VLANs therefore requires a Layer 3 device such as a router or Layer 3 switch. This process is called Inter-VLAN Routing.

Router-on-a-Stick

Router-on-a-Stick is a method of performing Inter-VLAN Routing using one physical router interface divided into multiple logical subinterfaces. For example, the physical interface G0/0 can have subinterfaces G0/0.10 and G0/0.20 for VLAN 10 and VLAN 20 respectively.

Subinterface

A subinterface is a logical interface created from a physical router interface. It allows one physical interface to handle traffic for multiple VLANs.

VLAN Tagging and 802.1Q

When multiple VLANs travel over a trunk, the receiving device needs to determine which VLAN an Ethernet frame belongs to. VLAN tagging provides this information. IEEE 802.1Q adds VLAN identification information to Ethernet frames carried across a trunk. The command encapsulation dot1Q 10 associates a router subinterface with VLAN 10.

Default Gateway

A default gateway is the device that a host sends packets to when the destination is outside its local network. In this experiment, the router subinterface acts as the default gateway for its corresponding VLAN. For example, 192.168.10.1 can be the gateway for VLAN 10.

DHCP

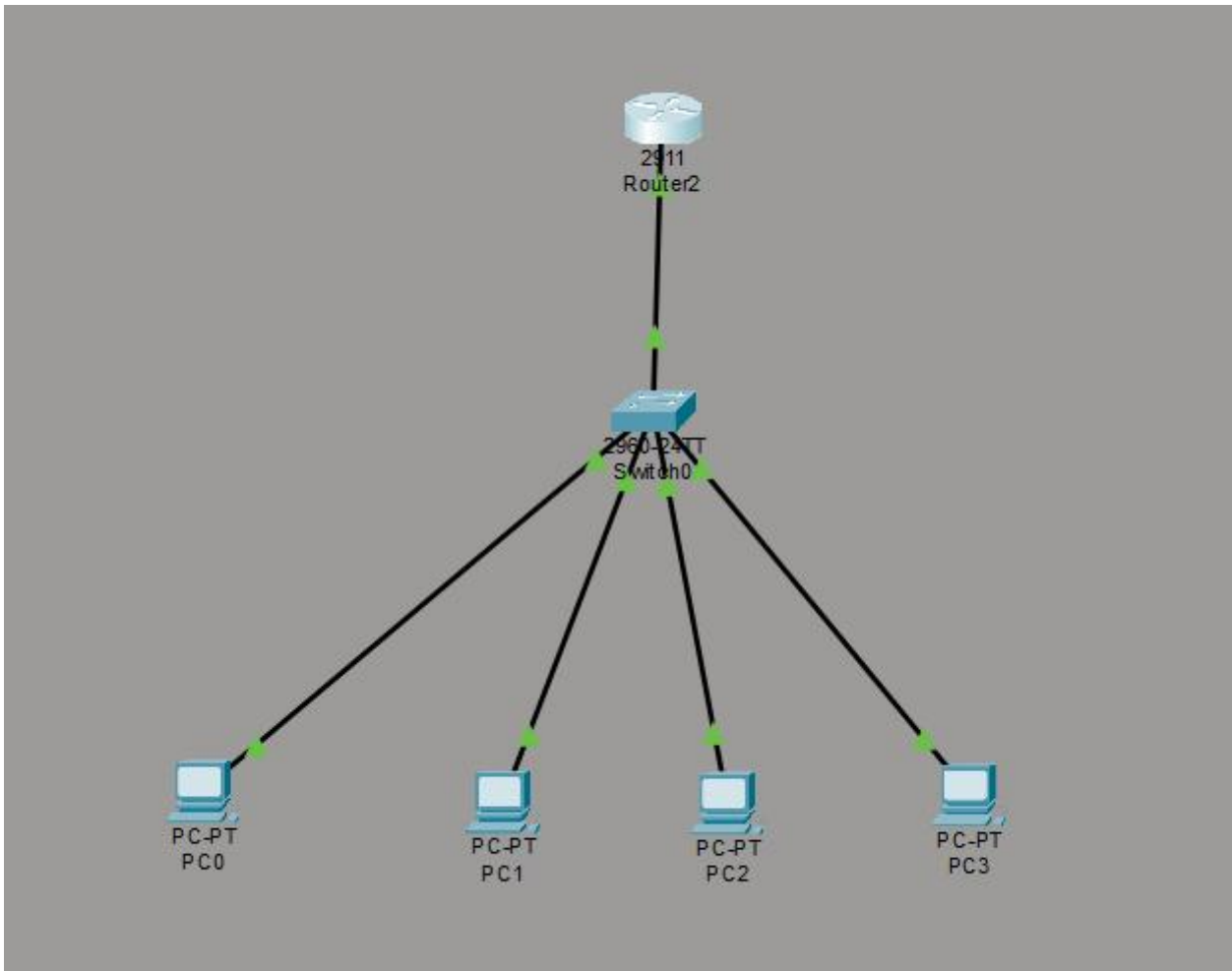
DHCP stands for Dynamic Host Configuration Protocol. DHCP automatically provides network configuration information to client devices, including IP address, subnet mask, default gateway, DNS server, and lease information. It reduces the need for manual configuration of each computer.

Router as DHCP Server

A Cisco router can be configured to act as a DHCP Server. In this experiment, a separate dedicated DHCP server is not required because the router performs both DHCP service and routing for multiple VLANs.

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Practical Steps:

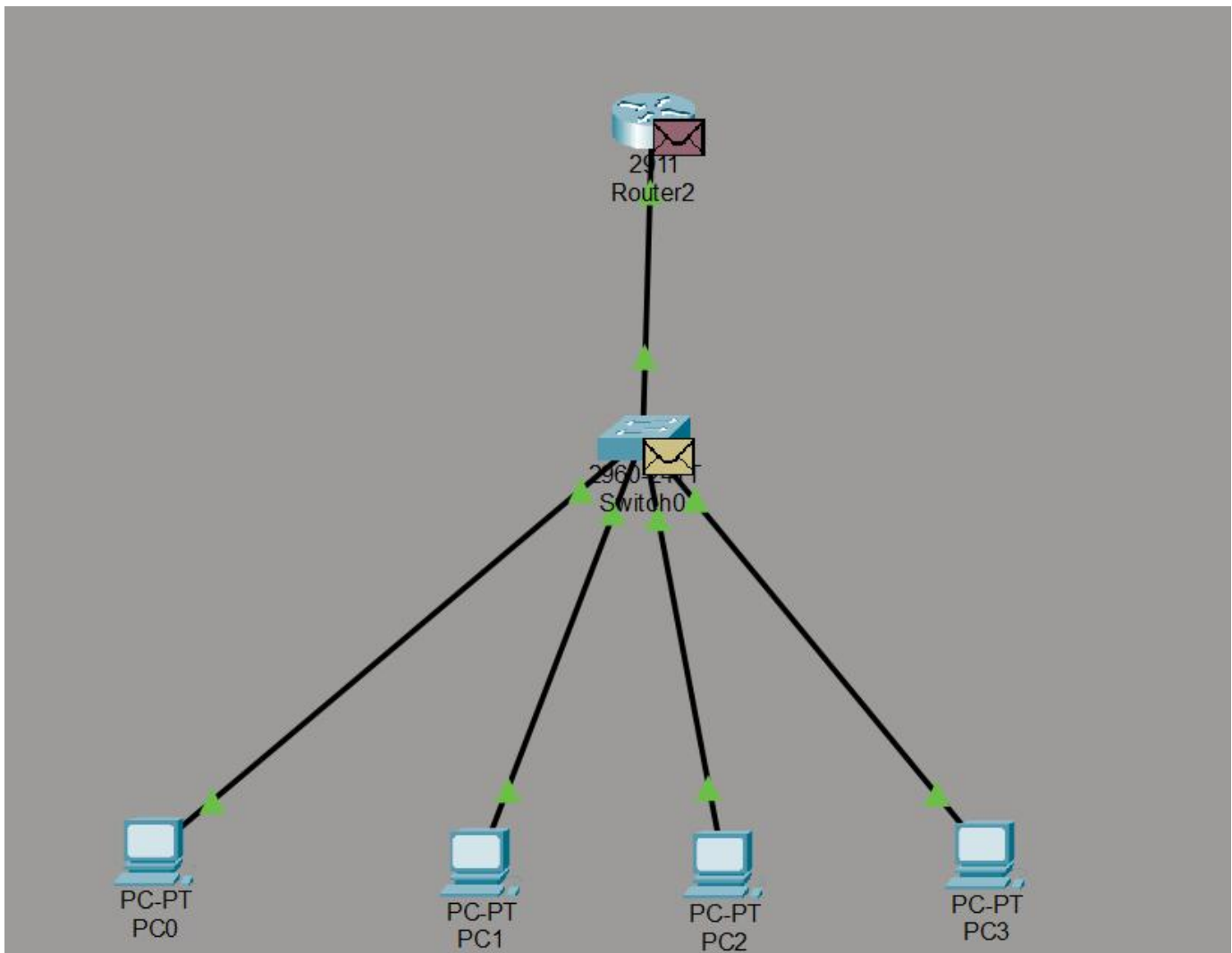


1. Open Cisco Packet Tracer.
2. Place the required devices in the workspace: one 2911 router, one 2960 switch, and four PCs.
3. Rename the router as R1 and the switch as S1. Rename the PCs as PC1, PC2, PC3, and PC4.
4. Connect the PCs to the switch and connect the switch to the router using Copper Straight-Through cables.
5. Configure the switch through the CLI and create the required VLANs, including VLAN 10 and VLAN 20.
6. Assign the appropriate switch ports to VLAN 10 and VLAN 20 as access ports.
7. Configure the switch port connected to the router as a trunk port.
8. Use show vlan brief to verify the VLAN assignments.
9. Use show interfaces trunk to verify that the router-facing interface, such as Fa0/24, is operating as a trunk.

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10. Configure the router interface connected to the switch and enable it.
11. Create router subinterface G0/0.10 for VLAN 10 and associate it using encapsulation dot1Q 10.
12. Assign 192.168.10.1 as the IP address of the VLAN 10 subinterface. This acts as the default gateway for VLAN 10.
13. Create router subinterface G0/0.20 for VLAN 20 and associate it using encapsulation dot1Q 20.
14. Assign 192.168.20.1 as the IP address of the VLAN 20 subinterface. This acts as the default gateway for VLAN 20.
15. Configure the router as a DHCP server and create a separate DHCP pool for VLAN 10.
16. Configure the VLAN 10 DHCP pool to provide addresses from the 192.168.10.x network and set 192.168.10.1 as the default gateway.
17. Create a separate DHCP pool for VLAN 20 to provide addresses from the 192.168.20.x network and set 192.168.20.1 as the default gateway.
18. Exclude the router gateway addresses from DHCP allocation so they are not assigned to PCs.
19. Configure each PC to obtain its IP configuration automatically using DHCP.
20. Verify that PCs in VLAN 10 receive 192.168.10.x addresses and PCs in VLAN 20 receive 192.168.20.x addresses.
21. Verify the router interfaces and DHCP configuration.
22. Test communication between PCs in the same VLAN and between different VLANs using the Ping command.
23. Observe that the router performs Inter-VLAN Routing and automatically provides IP configuration to clients through DHCP.

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Learning Outcome

1. Understand the concept of VLANs and their role in logically separating devices into different broadcast domains.
2. Develop practical skills in creating VLANs and assigning switch ports to appropriate VLANs in Cisco Packet Tracer.
3. Understand the difference between access ports and trunk ports and configure a trunk connection between a switch and router.
4. Understand Inter-VLAN Routing and the Router-on-a-Stick method using router subinterfaces.

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5. Configure router subinterfaces for multiple VLANs and assign appropriate default gateway addresses.
6. Configure a Cisco router as a DHCP server and create separate DHCP pools for different VLANs.
7. Automatically assign IP addresses, subnet masks, and default gateways to PCs using DHCP.
8. Verify VLAN, trunk, DHCP, and routing configurations and test communication between different VLANs.